Name: _____

SPREADSHEET TASK 1

Evaluating the modulus of a complex number 1

The modulus of a complex number is equal to the sum of the squares of the real and imaginary parts of a complex number.
We can use *Excel* to calculate the modulus of a complex number. Open up a new spreadsheet and enter in the following information.

	A	B	C	D	E
1	5	12	= COMPLEX (A1,B1)	= A1^2 + B1^2	= SQRT(D1)
2	3	−4			
3	−8	6			
4	−15	−8			
5					
6					
7					
.	.				
.	.				
.	.				

Copy the formulas in columns C, D and E down the columns.

Questions

1 Which column represents the real part of the complex number and which column represents the imaginary part?

2 Which column represents the modulus of a complex number?

3 Add five more complex numbers to your worksheet and find the modulus for each of them.

4 What limitations exist for using *Excel* to find the modulus of a complex number?

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Answers

1 Column A represents the real part and Column B represents the imaginary part.

2 Column E.

4 The modulus cannot be expressed as an exact number (a surd). If an exact answer is required, you would have to take the sum of the squares of the real and imaginary parts of the complex number (column D) and then express it as a surd, simplifying manually if possible.

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